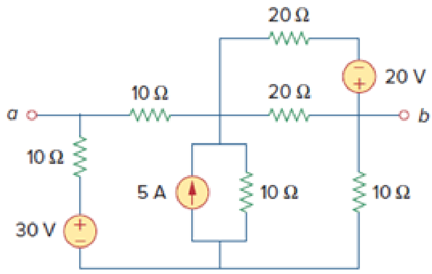
Chapter 4, Problem 42P

Problem

For the circuit in Fig. 4.109, find the Thevenin equivalent between terminals *a* and *b*.

Figure. 4.109:



Step-by-step solution

Step 1 of 10

Refer to the Figure 4.109 in the text book.

Draw the circuit diagram to determine the Thevenin voltage.

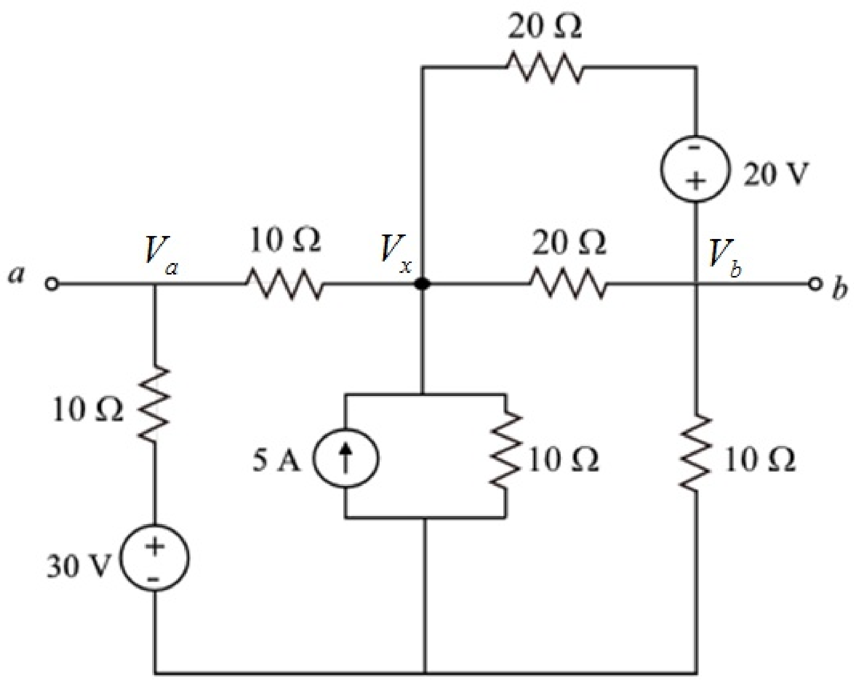


Figure 1

Step 2 of 10

Apply Kirchhoff's current law at node V_a.

$$\frac{V_a-30}{10}+\frac{V_a-V_x}{10}=0$$
$$\frac{V_a-30+V_a-V_x}{10}=0$$

$2V_a-V_x=30$ …… (1)

Apply Kirchhoff's current law at node V_b.

$$\frac{V_b-20-V_x}{20}+\frac{V_b}{10}+\frac{V_b-V_x}{20}=0$$
$$\frac{V_b-20-V_x+2V_b+V_b-V_x}{20}=0$$

$4V_b-2V_x=20$ …… (2)

Apply Kirchhoff's current law at node V_x.

$$\frac{V_x-V_a}{10}+\frac{V_x}{10}-5+\frac{V_x-V_b}{20}+\frac{V_x+20-V_b}{20}=0$$
$$\frac{2(V_x-V_a)+2V_x-100+(V_x-V_b)+(V_x+20-V_b)}{20}=0$$
$$2V_x-2V_a+2V_x-100+V_x-V_b+V_x+20-V_b=0$$

$-2V_a-2V_b+6V_x=80$ …… (3)

Step 3 of 10

Solve equations (1), (2) and (3).

$$V_a=30\,\text{V}$$
$$V_b=20\,\text{V}$$
$$V_x=30\,\text{V}$$

Determine the value of Thevenin voltage.

$$V_{TH}=V_a-V_b$$

Substitute $30\,\text{V}$ for V_a and $20\,\text{V}$ for V_b in the equation.

$$\begin{aligned}V_{TH}&=30\,\text{V}-20\,\text{V}\\&=10\,\text{V}\end{aligned}$$

Thus, the value of Thevenin voltage, V_{TH} is $\boxed{10\,\text{V}}$.

Step 4 of 10

Draw the circuit diagram to determine the Thevenin resistance.

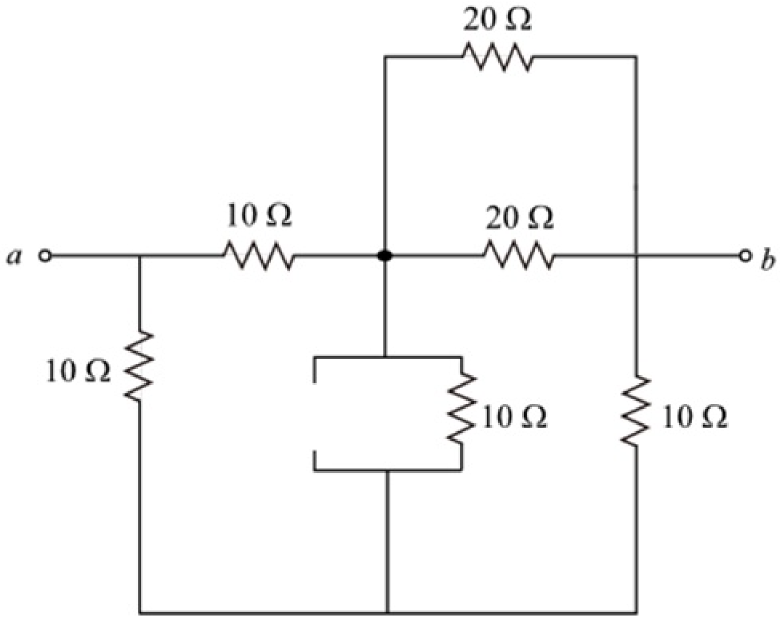


Figure 2

Step 5 of 10

In Figure 2, the resistors 20Ω and 20Ω are in parallel.

$$(20\Omega \parallel 20\ \Omega) = \frac{(20\Omega)(20\ \Omega)}{(20\Omega)+(20\ \Omega)}$$
$$= \frac{20\Omega}{2}$$
$$= 10\ \Omega$$

The simplified circuit diagram is shown in Figure 3.

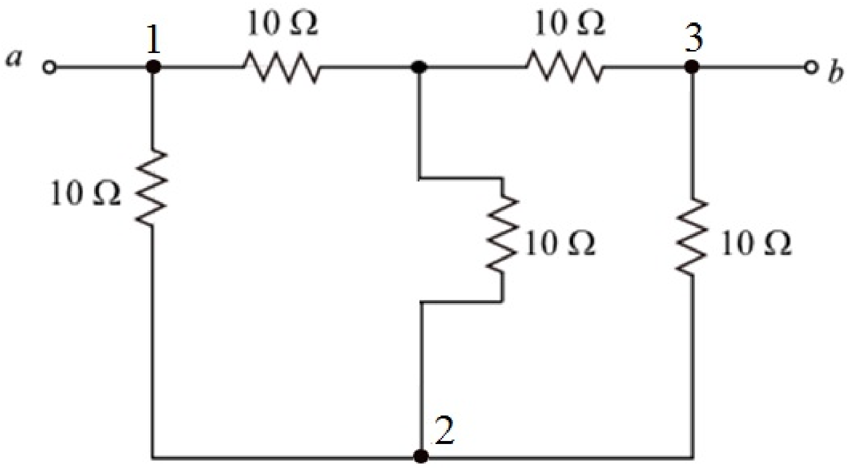


Figure 3

Step 6 of 10

Convert star network to delta network in Figure 3.

Find the resistance in between node-1 and node-2.

$$R_a = \frac{(10\Omega)(10\ \Omega)+(10\Omega)(10\ \Omega)+(10\Omega)(10\ \Omega)}{(10\ \Omega)}$$
$$= \frac{3(10\Omega)(10\ \Omega)}{(10\ \Omega)}$$
$$= 30\ \Omega$$

Find the resistance in between node-2 and node-3.

$$R_b = \frac{(10\Omega)(10\ \Omega)+(10\Omega)(10\ \Omega)+(10\Omega)(10\ \Omega)}{(10\ \Omega)}$$
$$= \frac{3(10\Omega)(10\ \Omega)}{(10\ \Omega)}$$
$$= 30\ \Omega$$

Find the resistance in between node-3 and node-1.

$$R_c = \frac{(10\Omega)(10\ \Omega)+(10\Omega)(10\ \Omega)+(10\Omega)(10\ \Omega)}{(10\ \Omega)}$$
$$= \frac{3(10\Omega)(10\ \Omega)}{(10\ \Omega)}$$
$$= 30\ \Omega$$

Step 7 of 10

The simplified circuit diagram is shown in Figure 4.

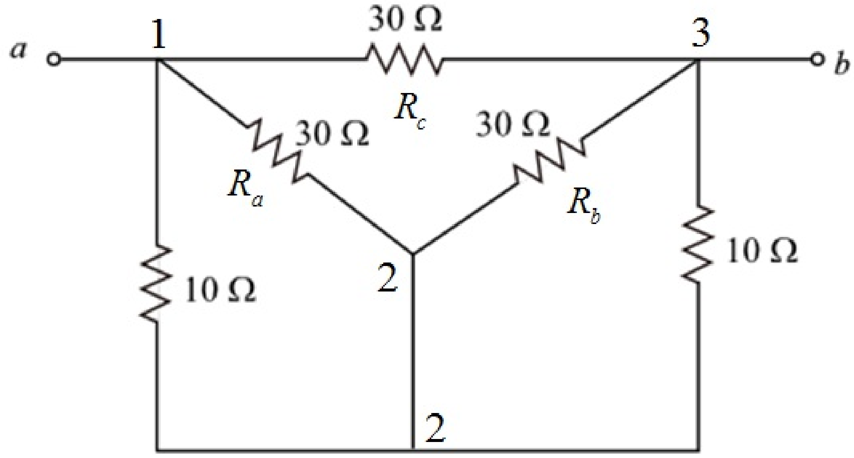


Figure 4

Step 8 of 10

In Figure 4, the resistors 10Ω and $30\ \Omega$ are in parallel.

$$10\Omega \,||\, 30\ \Omega = \frac{(10\Omega)(30\ \Omega)}{10\Omega + 30\ \Omega}$$
$$= 7.5\ \Omega$$

Draw the following modified circuit diagram.

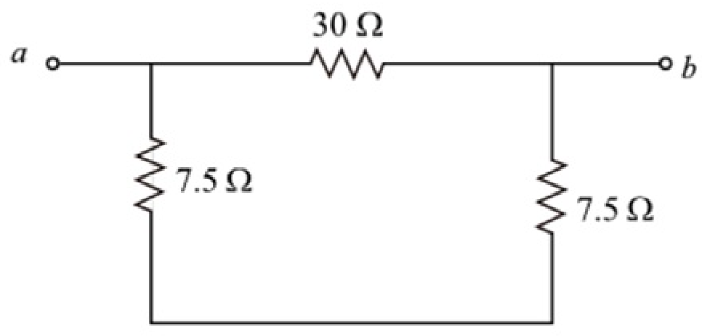


Figure 5

Step 9 of 10

Determine the value of Thevenin's resistance.

$$\begin{aligned} R_{TH} &= R_{ab} \\ &= (7.5\Omega + 7.5\Omega) \,||\, 30\Omega \\ &= 15\Omega \,||\, 30\Omega \\ &= \frac{(15\Omega)(30\Omega)}{15\Omega + 30\Omega} \end{aligned}$$

$$R_{TH} = 10\ \Omega$$

Thus, the value of Thevenin resistance, R_{TH} is $\boxed{10\Omega}$.

Step 10 of 10

Draw the Thevenin equivalent circuit.

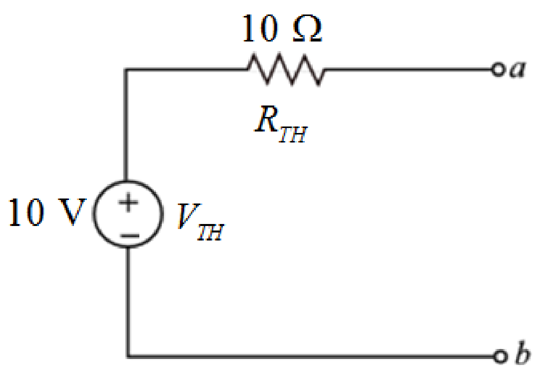


Figure 6

Thus, the required Thevenin equivalent circuit is sketched as shown in Figure 6.